

- 8 (a) momentum conservation hence momenta of photons are equal (but opposite) M1
 same momentum so same energy A1 [2]
- (b) (i) $(\Delta)E = (\Delta)mc^2$ C1
 $= 1.2 \times 10^{-28} \times (3.0 \times 10^8)^2$
 $= 1.08 \times 10^{-11} \text{ J}$ A1 [2]
- (ii) $E = hc / \lambda$
 $\lambda = (6.63 \times 10^{-34} \times 3.0 \times 10^8) / (1.08 \times 10^{-11})$ C1
 $= 1.84 \times 10^{-14} \text{ m}$ A1 [2]
- (iii) $\lambda = h / p$
 $p = (6.63 \times 10^{-34}) / (1.84 \times 10^{-14})$ C1
 $= 3.6 \times 10^{-20} \text{ N s}$ A1 [2]

Section B

- 9 (a) (i) point X shown correctly B1 [1]